Scale-invariant propagated class exclusion for direct semi-supervised clustering

Deshu Sun, Feng Wang*

College of Data Science and Information Engineering, Guizhou Minzu University, Guiyang, Guizhou 550025, P.R. China

*Corresponding author: wangf991@163.com, wangfeng@gzmu.edu.cn

Abstract

Semi-supervised non-negative matrix factorization (NMF) usually transfers labels by fitting latent coordinates or smoothing neighboring representations. These strategies specify what an unlabeled representation should resemble. We study the complementary problem of determining which class contributions should be suppressed. Propagated Class-Exclusion NMF (PCE-NMF) converts harmonic class evidence into sample-specific class-incompatibility weights and penalizes the corresponding rank-one contribution energies. The penalty and the induced direct clustering rule are invariant under the diagonal scaling ambiguity of NMF. We establish the harmonic-evidence properties, derive an equivalent canonical shrinkage model, and prove monotonic descent of the objective under the stated alternating updates. Experiments use six benchmark data sets, 20 paired random seeds, identical labeled splits, and evaluation restricted to unlabeled samples. Against eight baselines under a common permutation-aligned clustering protocol, PCE-NMF obtains the highest mean accuracy and normalized mutual information on COIL20 and COIL100 and the highest accuracy on ORL. Its native class-semantic decisions closely match the aligned scores on all six data sets. The controlled ablation shows substantial gains over hard class exclusion in every paired trial. A harmonic-propagation-only control separates propagation from its coupling with factorization: PCE-NMF improves both metrics on COIL20 and COIL100, while propagation alone is stronger on YaleB and PIE. These results identify propagated class exclusion as an effective direct-clustering mechanism with its clearest additional gains on multi-class object data. Code and results are available at https://github.com/matrix-research-lab-Wf/PCE-NMF.

Keywords: non-negative matrix factorization; semi-supervised clustering; label propagation; class exclusion; direct clustering; scale invariance

1 Introduction

Non-negative matrix factorization (NMF) has become a widely used tool for low-dimensional representation and clustering because its non-negativity constraints lead to an additive and often interpretable representation of non-negative data [1, 2]. Given a non-negative data matrix $X \in \mathbb{R}_+^{m\times n}$, standard NMF seeks two non-negative factor matrices whose product approximates X. The learned coefficient matrix can reveal latent clustering structure, while standard NMF ignores available class labels.

Semi-supervised NMF incorporates partial labels into the factorization. Existing methods mainly follow several representative strategies. Constraint-based methods enforce label-consistent structures in the latent representation [3]. Discriminative methods impose penalties that encourage labeled samples to exhibit class-consistent representations [4, 5]. More recently, direct constraint models avoid fitting a prescribed label vector. For example, GOCNMF fixes labeled representations to one-hot vectors and transfers this structure through graph regularization [6], whereas LCNMF sets only class-inconsistent coordinates of labeled representations to zero and leaves their class-consistent coordinates adaptive [7]. Another line of research propagates labels or pairwise constraints through sample graphs. Constraint-propagation NMF modifies pairwise relations [8]; HSSNMF propagates constraints through a hypergraph [9]; SPNMF jointly propagates pointwise and pairwise supervision [10]; and weighted or high-order label propagation regularizes latent representations or label-prediction variables [11, 12, 13]. Self-supervised and robust semi-supervised NMF models further enrich the supervisory signal or the reconstruction loss [14]. Across these formulations, supervision specifies target coordinates, sample relations, or neighborhood agreement.

This work considers a different question. Instead of asking whether an unlabeled representation should be close to a propagated label vector, we ask whether propagated class evidence can be used to suppress *class-incompatible NMF contributions*. Let P_{ik} denote the propagated evidence that sample i belongs to class k. A large value of P_{ik} indicates that the kth class contribution should be retained, whereas a small value indicates incompatibility and should induce stronger suppression. To implement this principle without introducing dependence on the arbitrary column scaling of NMF, we use the scale-invariant contribution energy

$$\|u_k\|_2^2 v_{ik}^2,$$

where u_k is the kth basis vector and v_{ik} is the corresponding coefficient of sample i.

Based on this observation, we propose *Propagated Class-Exclusion Non-negative Matrix Factorization* (PCE-NMF). Harmonic propagation first produces a class-evidence matrix from the available labels. Its complement defines a sample-specific class-incompatibility field. PCE-NMF then penalizes the contribution energy of each class according to this incompatibility, while labeled samples retain an explicit hard exclusion of their class-inconsistent coordinates. Supervision is expressed as class-dependent shrinkage, distinct from absolute one-hot fitting and pairwise graph smoothing.

The propagated penalty respects the standard diagonal s-

caling ambiguity of NMF. Indeed, under the transformation $U \mapsto US$ and $V \mapsto VS^{-1}$ with a positive diagonal matrix S, both the reconstruction and every supervised contribution remain unchanged. This property leads to a canonical interpretation of PCE-NMF: after normalizing each basis column, the proposed regularizer becomes a sample- and class-dependent quadratic shrinkage on the canonical coefficient matrix. The supervised quantity is therefore independent of arbitrary positive diagonal rescaling of the factors.

The main contributions of this work are summarized as follows.

1. We introduce propagated class exclusion, which converts harmonic class evidence into sample-specific suppression of class-incompatible rank-one contributions.
2. We prove the harmonic-evidence bounds and diagonal-scale invariance, and derive an equivalent canonical shrinkage representation.
3. We derive multiplicative updates with monotonic objective descent and a direct class-energy decision rule built from the same invariant quantity.
4. Paired experiments separate the contribution of propagated exclusion from hard support constraints and characterize its behavior across label rates, graph neighborhoods, data sets, and competing NMF mechanisms.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the proposed PCE-NMF model and its direct clustering rule. Section 4 presents the theoretical analysis, optimization algorithm, and computational complexity. Section 5 reports the experimental results, and Section 6 concludes the paper.

2 Related Work

2.1 Label-constrained and discriminative NMF

A direct way of incorporating partial labels into NMF is to constrain the latent representation. Constrained NMF (CNMF) constructs a label-constraint matrix so that labeled samples from the same class share related low-dimensional representations [3]. Subsequent methods replace the exact structural constraint with soft discriminative penalties. For example, discriminative semi-supervised NMF methods encourage the representations of labeled samples to be consistent with their known labels while retaining reconstruction flexibility [4]. More recent direct-clustering formulations further weaken the requirement that a labeled representation must match a prescribed one-hot vector. EDNMF constructs discriminative regularizers from the relative magnitudes of class-consistent and class-inconsistent elements of labeled representations [5]. LCNMF instead fixes the class-inconsistent coordinates of each labeled representation to zero while leaving its class-consistent coordinate as a learnable variable [7]. These formulations improve representation flexibility and allow cluster assignments to be read directly from the learned coefficients.

GOCNMF adopts a different direct strategy by fixing labeled representations to their one-hot label vectors and using graph regularization to guide the unlabeled representations [6]. If the labeled samples are placed first, its representation matrix has the form

$$\widetilde{V} = \begin{bmatrix} C \\ V_{\mathcal{U}} \end{bmatrix},$$

where C is the one-hot label matrix. The graph term then encourages neighboring samples to have similar latent representations. Such a design provides strong class semantics, but the supervision is imposed directly on the latent coordinates.

The proposed PCE-NMF differs from these approaches in the object on which supervision is imposed. For labeled samples, PCE-NMF retains a hard class-exclusion structure similar in spirit to local coordinate constraints. For unlabeled samples, propagated supervision acts on the scale-invariant class contribution $\|u_k\|_2^2 v_{ik}^2$ through a sample-specific class-incompatibility weight.

2.2 Propagation-based semi-supervised NMF

Label propagation provides an effective mechanism for transferring limited supervision from labeled to unlabeled samples. Several semi-supervised NMF methods have therefore coupled matrix factorization with label propagation, pairwise-constraint propagation, or adaptive graph learning [8, 11, 10]. Recent Pattern Recognition models illustrate complementary propagation routes. HSSNMF propagates pairwise constraints through a hypergraph and uses the resulting similarity structure in symmetric NMF [9]. SPNMF propagates both labels and pairwise constraints and combines them with pivotal-aware weighting [10]. CSNMF addresses a different axis by combining pointwise and pairwise supervision with a correntropy loss for contaminated data [14]. These models expand the amount, order, or robustness of the supervisory information used by NMF.

SNMFWLP integrates weighted label propagation directly into NMF and uses the low-dimensional representation itself as a label-prediction variable [12]. Its representative objective contains both a graph smoothness term and a weighted fitting term,

$$\begin{aligned} &\|X - WH^T\|_F^2 + \alpha \operatorname{Tr}(H^T L H) \\ &\quad + \beta \operatorname{Tr}\big((H-M)^T \mathcal{U} (H-M)\big), \end{aligned}$$

together with an orthogonality constraint on the basis matrix. Thus, propagated supervision explicitly pulls the learned representation toward a weighted label target.

Other recent models enrich the propagation mechanism itself. Positive and negative label propagation can be combined with constraint propagation to refine neighborhood relations [11], while high-order propagation methods exploit multi-hop relations beyond first-order graph neighbors [13]. These developments demonstrate that propagation can substantially enhance semi-supervised NMF, but their supervision is still primarily expressed through representation fitting, pairwise relations, or propagation-induced graph regularization.

PCE-NMF uses propagation differently. The harmonic solution is computed before factorization and is converted into a fixed class-incompatibility field. The resulting penalty is

diagonal with respect to sample-class coordinates:

$$\sum_{i,k}(1-P_{ik})\|u_k\|_2^2 v_{ik}^2.$$

Hence, unlike a Laplacian term, it introduces no cross-sample product $v_i^T v_j$. Unlike label fitting, it does not force v_i to approximate a prescribed vector. Its role is instead to attenuate class contributions that are inconsistent with the propagated evidence.

2.3 Position of the proposed method

The distinction among representative semi-supervised NMF strategies can be summarized by the location and form of supervision. Label-fitting methods specify desirable latent coordinates, discriminative methods compare class-consistent and class-inconsistent coordinates, and graph-based methods couple representations of neighboring samples. PCE-NMF occupies a distinct mechanism-level position: propagated evidence determines the strength of a class-wise exclusion penalty on scale-invariant contribution energies.

Standard NMF admits the diagonal transformation

$$U \mapsto US, \qquad V \mapsto VS^{-1},$$

for any positive diagonal S. A supervisory quantity based solely on a raw coefficient may therefore depend on an arbitrary factor scaling unless an additional normalization or structural constraint is imposed. In PCE-NMF, the product $\|u_k\|_2^2 v_{ik}^2$ is invariant under this transformation. The proposed supervision consequently has a direct interpretation that is independent of the NMF column gauge.

3 Propagated Class-Exclusion NMF

Figure 1 summarizes the two coupled routes of PCE-NMF. The propagation route converts partial labels and the cosine p-nearest-neighbor graph into harmonic class evidence P and class-exclusion weights Q. The factorization route learns U and V under hard labeled-support constraints. The two routes are coupled through the class-specific contribution energy $s_{ik} = \|u_k\|_2^2 v_{ik}^2$, which is regularized by q_{ik} and used directly for cluster assignment.

3.1 Notation and standard NMF

Let

$$X = [x_1, \ldots, x_n] \in \mathbb{R}_+^{m\times n}$$

contain n non-negative samples from c classes. Let $\mathcal{L}$ and $\mathcal{U}$ denote the labeled and unlabeled index sets, respectively, with $\mathcal{L}\cap\mathcal{U} = \varnothing$ and $\mathcal{L}\cup\mathcal{U} = \{1,\ldots,n\}$.

We consider a rank-c factorization

$$X \approx UV^T,$$

where

$$U = [u_1, \ldots, u_c] \in \mathbb{R}_+^{m\times c}, \qquad V = [v_{ik}] \in \mathbb{R}_+^{n\times c}.$$

Standard Euclidean NMF minimizes

$$\min_{U,V\geq 0} \|X - UV^T\|_F^2. \tag{1}$$

For a labeled sample $i \in \mathcal{L}$, let $y_i \in \{1,\ldots,c\}$ denote its class label. Let

$$C \in \{0,1\}^{|\mathcal{L}|\times c}$$

be the corresponding one-hot label matrix.

3.2 Harmonic class evidence

All samples are normalized to unit Euclidean norm before graph construction. For each sample x_i, let $\mathcal{N}_p(i)$ contain the p other samples x_j, $j \neq i$, with the largest cosine similarities $x_i^T x_j$. The self-similarity is excluded before neighbor selection. Ties are resolved by the ascending sample index returned by MATLAB's stable ordering. The binary union graph is

$$W_{ij} = \begin{cases} 1, & j \in \mathcal{N}_p(i) \text{ or } i \in \mathcal{N}_p(j), \\ 0, & \text{otherwise}, \end{cases} \qquad W_{ii} = 0. \tag{2}$$

Thus $W \in \mathbb{R}_+^{n\times n}$ is symmetric and contains no self-loop. Let

$$D = \operatorname{diag}(W\mathbf{1}), \qquad L = D - W$$

be its degree matrix and graph Laplacian. Class evidence is obtained from the Gaussian-field harmonic solution [15].

Because disconnected components without labeled samples cannot receive reliable class evidence, harmonic propagation is performed separately on each connected component. Consider a component containing at least one labeled node, and partition its vertices into labeled and unlabeled sets. The propagated class-evidence matrix is defined by

$$P_{\mathcal{L}} = C, \qquad L_{\mathcal{U}\mathcal{U}} P_{\mathcal{U}} = W_{\mathcal{U}\mathcal{L}} C. \tag{3}$$

The implementation uses a fixed numerical stabilization $\delta = 10^{-12}$. For the unlabeled vertices of each anchored component, it computes

$$Z = (L_{\mathcal{U}\mathcal{U}} + \delta I)^{-1} W_{\mathcal{U}\mathcal{L}} C, \qquad P_{ik} = \frac{[Z_{ik}]_+}{\sum_{r=1}^{c}[Z_{ir}]_+}, \tag{4}$$

where $[z]_+ = \max\{z, 0\}$. The denominator is positive on every anchored component. Equation (3) defines the harmonic field, and Eq. (4) defines the canonical finite-precision map used by the released implementation.

For every unlabeled sample contained in an anchored component, P_{ik} represents the propagated evidence of class k. Let $\mathcal{A} \subseteq \mathcal{U}$ denote the set of such anchored unlabeled samples. We define the class-incompatibility field

$$q_{ik} = \begin{cases} 1 - P_{ik}, & i \in \mathcal{A}, \\ 0, & i \in \mathcal{L} \text{ or } i \in \mathcal{U}\setminus\mathcal{A}. \end{cases} \tag{5}$$

Thus, high class evidence yields weak exclusion, whereas low class evidence yields strong exclusion. Setting $q_{ik} = 0$ on unanchored components avoids introducing artificial supervision when no labeled information can reach the sample.

3.3 Hard exclusion for labeled samples

For each labeled sample, we impose

$$v_{ik} = 0, \qquad i \in \mathcal{L}, \quad k \neq y_i. \tag{6}$$

The class-consistent coefficient $v_{i\,y_i}$ remains learnable. The known label fixes the support pattern of the labeled representation, while the factorization determines its amplitude.

Table 1: Mechanisms and native assignment rules of the nine methods in the formal comparison. Class-semantic representations have coordinates tied to the c classes. The assignment rule is the one used by each method in the aligned experiments.

Method	Supervision mechanism	Graph/propagation	Class-semantic representation	Native assignment
NMF	none	none	no	50-start K-means
CNMF	constrained coefficient structure	none	no	50-start K-means
GNMFLD	labeled-coordinate fitting	graph Laplacian	yes	row-wise maximum
EDNMF (ERDNMF)	element-ratio discrimination	none	yes	row-wise maximum
LCNMF	labeled wrong-class suppression	none	yes	row-wise maximum
GOCNMF	one-hot labeled representation	graph smoothing	yes	row-wise maximum
SNMFWLP	weighted label fitting	label propagation	no	50-start K-means
HLPNMF	high-order label propagation	hypergraph propagation	yes	row-wise maximum
PCE-NMF	propagated class incompatibility	harmonic propagation	yes	$\arg\max_k \|u_k\|_2^2 v_{ik}^2$

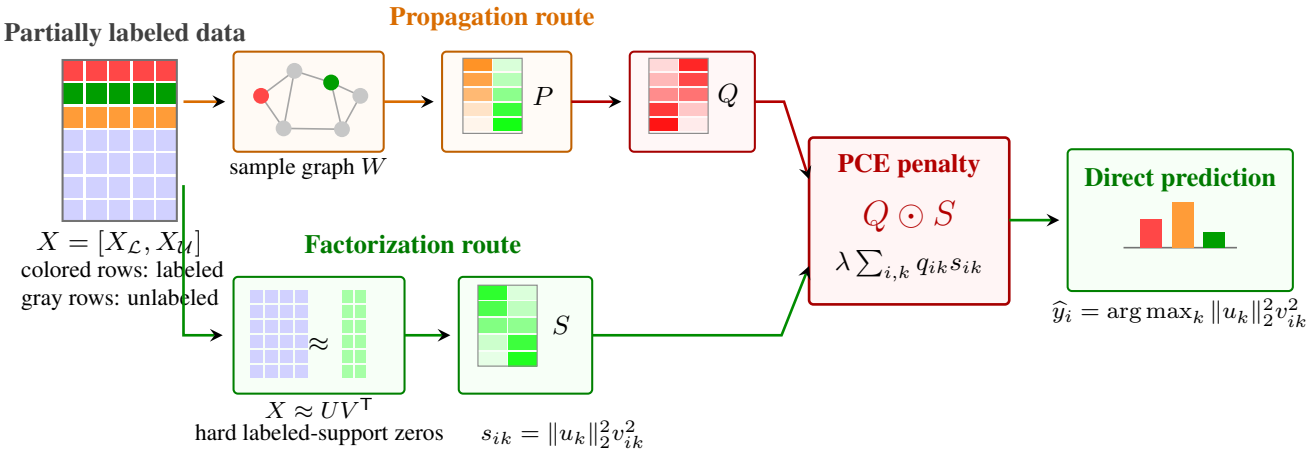


Figure 1: Overview of PCE-NMF. Partial labels are propagated within graph components to obtain harmonic class evidence. The evidence is converted into sample-specific class-exclusion weights, which suppress incompatible rank-one reconstruction contributions. Clusters are assigned directly using the scale-invariant contribution energy.

3.4 The PCE-NMF model

For sample i and class k, define the class-specific contribution energy

$$s_{ik} = \|u_k\|_2^2 v_{ik}^2. \tag{7}$$

This quantity is the squared Frobenius norm of the rank-one term $u_k v_{ik}$ in the reconstruction of sample i. The complete reconstruction energy also contains cross-products between different rank-one terms; the score in (7) measures the magnitude of one class contribution. The proposed PCE-NMF model is

$$\min_{U,V\geq 0} J(U,V) = \|X - UV^T\|_F^2 + \lambda \sum_{i=1}^{n}\sum_{k=1}^{c} q_{ik}\|u_k\|_2^2 v_{ik}^2, \tag{8}$$

subject to (6), where $\lambda > 0$ controls the overall strength of propagated exclusion.

The second term in (8) has a direct interpretation. If P_{ik} is large, then $q_{ik} = 1 - P_{ik}$ is small and the kth contribution of sample i is weakly penalized. If P_{ik} is small, the same contribution receives stronger shrinkage. Propagated evidence therefore controls the retained strength of each class-specific NMF contribution.

When $\lambda = 0$, (8) reduces to Hard Class-Exclusion NMF (HCE-NMF), which retains the hard support constraint (6) and removes propagated exclusion.

3.5 Direct clustering

Because the score in (7) measures a class-specific contribution in a scale-invariant form, PCE-NMF produces cluster assignments directly:

$$\widehat{y}_i = \arg\max_{1\leq k\leq c} \|u_k\|_2^2 v_{ik}^2. \tag{9}$$

4 Theoretical Analysis

We next establish the main structural properties of PCE-NMF and derive its multiplicative optimization scheme.

4.1 Validity of harmonic class evidence

Lemma 1 (Validity of harmonic class evidence). Consider a connected graph component containing at least one labeled vertex. Both the harmonic solution in (3) and the stabilized normalized field in (4) satisfy

$$P_{\mathcal{U}} \geq 0, \qquad P_{\mathcal{U}}\mathbf{1}_c = \mathbf{1}_{|\mathcal{U}|}.$$

Consequently,

$$0 \leq P_{ik} \leq 1$$

for every unlabeled vertex in the component, and hence $0 \leq q_{ik} \leq 1$.

Proof. For a connected component containing at least one labeled vertex, the Dirichlet Laplacian $L_{\mathcal{U}\mathcal{U}}$ is a nonsingular M-matrix. Therefore

$$L_{\mathcal{U}\mathcal{U}}^{-1} \geq 0.$$

Since $W_{\mathcal{U}\mathcal{L}}C \geq 0$, equation (3) gives

$$P_{\mathcal{U}} = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}} C \geq 0.$$

Moreover, because the rows of the full graph Laplacian sum to zero,

$$L_{\mathcal{U}\mathcal{U}}\mathbf{1}_{|\mathcal{U}|} = W_{\mathcal{U}\mathcal{L}}\mathbf{1}_{|\mathcal{L}|}.$$

Since $C\mathbf{1}_c = \mathbf{1}_{|\mathcal{L}|}$,

$$P_{\mathcal{U}}\mathbf{1}_c = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}} C\mathbf{1}_c = L_{\mathcal{U}\mathcal{U}}^{-1} W_{\mathcal{U}\mathcal{L}}\mathbf{1}_{|\mathcal{L}|} = \mathbf{1}_{|\mathcal{U}|}.$$

Non-negativity and unit row sums imply $0 \leq P_{ik} \leq 1$. The result for q_{ik} follows from (5).

For the implemented field, $L_{\mathcal{U}\mathcal{U}} + \delta I$ is also a nonsingular M-matrix, so $Z \geq 0$. Every unlabeled block in an anchored component has a path to a labeled vertex, which gives a positive row sum in Z. The row normalization in (4) therefore yields a non-negative vector with unit sum for every anchored unlabeled sample. The same bounds on P_{ik} and q_{ik} follow. □

4.2 Scale invariance and canonical representation

Theorem 1 (Scale invariance)**.** *Let*

$$S = \mathrm{diag}(s_1, \ldots, s_c), \qquad s_k > 0,$$

and define

$$\widetilde{U} = US, \qquad \widetilde{V} = VS^{-1}.$$

Then

$$J(\widetilde{U}, \widetilde{V}) = J(U, V).$$

Furthermore, the hard-support constraints in (6) *are preserved.*

Proof. The reconstruction is unchanged because

$$\widetilde{U}\widetilde{V}^T = US(VS^{-1})^T = UV^T.$$

For each class k,

$$\|\widetilde{u}_k\|_2^2 \widetilde{v}_{ik}^2 = s_k^2 \|u_k\|_2^2 \frac{v_{ik}^2}{s_k^2} = \|u_k\|_2^2 v_{ik}^2.$$

Hence every term of the propagated-exclusion penalty is unchanged. Finally, $v_{ik} = 0$ if and only if $\widetilde{v}_{ik} = 0$, so the feasible support pattern is preserved. □

Corollary 1 (Scale-invariant direct decision)**.** *Under the transformation in Theorem 1,*

$$\arg\max_k \|\widetilde{u}_k\|_2^2 \widetilde{v}_{ik}^2 = \arg\max_k \|u_k\|_2^2 v_{ik}^2.$$

Therefore, the direct clustering rule (9) *is independent of the standard diagonal scaling ambiguity of NMF.*

Proposition 1 (Canonical shrinkage representation)**.** *Assume $\|u_k\|_2 > 0$ for $k = 1, \ldots, c$. Define*

$$d_k = \|u_k\|_2, \qquad D_u = \mathrm{diag}(d_1, \ldots, d_c),$$

and

$$B = UD_u^{-1}, \qquad Z = VD_u.$$

Then $\|b_k\|_2 = 1$, $UV^T = BZ^T$, and PCE-NMF is equivalently written as

$$\min_{B,Z\geq 0} \|X - BZ^T\|_F^2 + \lambda \sum_{i=1}^{n}\sum_{k=1}^{c} q_{ik} z_{ik}^2, \tag{10}$$

subject to $\|b_k\|_2 = 1$ and the corresponding hard-support constraints.

Proof. By construction,

$$\|b_k\|_2 = \frac{\|u_k\|_2}{d_k} = 1,$$

and

$$BZ^T = UD_u^{-1}(VD_u)^T = UV^T.$$

Moreover,

$$z_{ik} = d_k v_{ik},$$

so

$$z_{ik}^2 = \|u_k\|_2^2 v_{ik}^2.$$

Substituting these identities into (8) gives (10). □

Proposition 1 shows that the propagated-exclusion term can be interpreted as a sample- and class-dependent quadratic shrinkage on a canonical coefficient representation. Its class semantics are invariant to arbitrary column scaling of the factorization. The canonical form is defined on nondegenerate factor pairs whose basis columns are all nonzero.

4.3 Evidence-exclusion relation

For an anchored unlabeled sample i, define

$$s_i = (s_{i1}, \ldots, s_{ic})^T, \qquad S_i = \sum_{k=1}^{c} s_{ik},$$

and let

$$R_i = \sum_{k=1}^{c} (1 - P_{ik}) s_{ik}$$

denote its propagated-exclusion penalty before multiplication by λ.

Proposition 2 (Evidence-exclusion decomposition)**.** *For every anchored unlabeled sample,*

$$R_i = S_i - P_i^T s_i. \tag{11}$$

Moreover, if

$$P_i^{\max} = \max_k P_{ik}, \qquad P_i^{\min} = \min_k P_{ik},$$

then

$$(1 - P_i^{\max}) S_i \leq R_i \leq (1 - P_i^{\min}) S_i. \tag{12}$$

Proof. Using $\sum_k P_{ik} = 1$ from Lemma 1,

$$R_i = \sum_k (1 - P_{ik}) s_{ik} = \sum_k s_{ik} - \sum_k P_{ik} s_{ik}$$
$$= S_i - P_i^T s_i.$$

Because $s_{ik} \geq 0$,

$$P_i^{\min} S_i \leq P_i^T s_i \leq P_i^{\max} S_i.$$

Subtracting these inequalities from S_i gives (12). □

For fixed total contribution energy S_i, minimizing R_i is therefore equivalent to maximizing $P_i^T s_i$. This provides an interpretation of PCE-NMF as aligning class-specific contribution energy with propagated class evidence without explicitly fitting s_i to P_i.

Corollary 2 (Uniform-evidence neutrality)**.** *If*

$$P_{ik} = \frac{1}{c}, \qquad k = 1, \ldots, c,$$

then

$$R_i = \left(1 - \frac{1}{c}\right) S_i.$$

Thus, uniform propagated evidence induces class-independent shrinkage of the total contribution energy.

4.4 Multiplicative updating rules

Let

$$a_k = \|u_k\|_2^2, \qquad b_k = \sum_{i=1}^{n} q_{ik} v_{ik}^2.$$

The gradient of (8) with respect to U is

$$\nabla_U J = 2U(V^T V) - 2XV + 2\lambda U \operatorname{diag}(b_1, \ldots, b_c).$$

Separating positive and negative parts gives the multiplicative rule

$$U \leftarrow U \odot \frac{XV}{U(V^T V) + \lambda U \operatorname{diag}(b_1, \ldots, b_c)}. \tag{13}$$

Similarly,

$$\nabla_V J = 2V(U^T U) - 2X^T U + 2\lambda (Q \odot V) \operatorname{diag}(a_1, \ldots, a_c),$$

where $Q = [q_{ik}]$. Hence every free coefficient is updated by

$$V \leftarrow V \odot \frac{X^T U}{V(U^T U) + \lambda (Q \odot V) \operatorname{diag}(a_1, \ldots, a_c)}. \tag{14}$$

After each V update, the constrained entries in (6) remain fixed at zero.

Numerical implementation. The MATLAB R2019a implementation initializes every free entry by a positive common factor plus 10^{-2} and evaluates each multiplicative denominator as $\max\{d, 10^{-12}\}$ elementwise. After an update, free entries are clipped below at 10^{-12}, and labeled wrong-class entries are reset exactly to zero. The graph and harmonic evidence remain fixed during factorization. All reported experiments use 50 PCE-NMF iterations. These operations specify the executed finite-precision algorithm; the theorem below analyzes the corresponding ideal multiplicative updates.

The stabilized harmonic map was checked against the unshifted Dirichlet solve on all 120 independent PCE-NMF trials. The maximum entrywise difference in the exclusion field was 2.54×10^{-11}; all sample predictions were identical, and the ACC and NMI differences were exactly zero at machine precision. The stabilized map in Eq. (4) is used as the canonical released implementation, and the audit records are included in the public repository.

Proposition 3 (Evidence-monotone one-step shrinkage)**.** *Fix the current U and V, and consider an anchored unlabeled entry $v_{ik} > 0$ for which the numerator and denominator in* (14) *are strictly positive. In one application of* (14)*, the updated value v_{ik}^+ is a non-decreasing function of P_{ik}.*

Proof. For fixed current factors, (14) can be written as

$$v_{ik}^+ = v_{ik} \frac{(X^T U)_{ik}}{[V(U^T U)]_{ik} + \lambda(1 - P_{ik}) \|u_k\|_2^2 v_{ik}}.$$

Let

$$A = [V(U^T U)]_{ik}, \quad B = \|u_k\|_2^2 v_{ik}, \quad C = v_{ik}(X^T U)_{ik}.$$

Since $A, B, C \geq 0$,

$$\frac{\partial v_{ik}^+}{\partial P_{ik}} = \frac{\lambda BC}{[A + \lambda(1 - P_{ik})B]^2} \geq 0.$$

Thus, greater propagated evidence results in weaker one-step shrinkage of the corresponding class coefficient. □

Proposition 3 characterizes the immediate response of one coefficient update to its propagated class evidence at fixed factors.

4.5 Monotonicity of the objective

We use the following standard majorization fact.

Lemma 2. Let A be symmetric and entrywise non-negative, and let $\bar{z} > 0$. Then

$$K(\bar{z}) = \operatorname{diag}\left(\frac{(A\bar{z})_1}{\bar{z}_1}, \ldots, \frac{(A\bar{z})_r}{\bar{z}_r}\right)$$

satisfies

$$K(\bar{z}) - A \succeq 0.$$

Proof. For any $x \in \mathbb{R}^r$,

$$x^T (K(\bar{z}) - A)\, x = \frac{1}{2} \sum_{p,q} A_{pq} \left(x_p \sqrt{\frac{\bar{z}_q}{\bar{z}_p}} - x_q \sqrt{\frac{\bar{z}_p}{\bar{z}_q}} \right)^2$$
$$\geq 0.$$

This completes the proof. □

Theorem 2 (Monotonic decrease of PCE-NMF)**.** *Assume that, at every iteration, all free entries and all numerators and denominators in the ideal multiplicative updates* (13)–(14) *are strictly positive. Apply these updates in exact arithmetic while keeping the hard-support zeros fixed. Then*

$$J(U^{(t+1)}, V^{(t+1)}) \leq J(U^{(t)}, V^{(t)}).$$

Consequently, the sequence of objective values converges.

Proof. First fix V. Let

$$b_k = \sum_i q_{ik} v_{ik}^2$$

and define

$$A_U = V^T V + \lambda \operatorname{diag}(b_1, \ldots, b_c).$$

Up to terms independent of U,

$$J(U, V) = \operatorname{Tr}(U A_U U^T) - 2 \operatorname{Tr}(XVU^T) + \text{const}.$$

The matrix A_U is symmetric and entrywise non-negative. Applying Lemma 2 row by row yields the standard diagonal auxiliary-function majorizer for this non-negative quadratic problem. Minimizing the majorizer gives (13). Hence the U update does not increase J.

Next fix U. For sample i, define

$$A_i = U^T U + \lambda \operatorname{diag}\big(q_{i1} a_1, \ldots, q_{ic} a_c\big), \quad a_k = \|u_k\|_2^2.$$

Up to constants independent of v_i,

$$J_i(v_i) = v_i^T A_i v_i - 2 x_i^T U v_i.$$

Again, A_i is symmetric and entrywise non-negative. Restricting to the free coordinates of v_i, its corresponding principal submatrix retains these properties. Lemma 2 therefore gives a separable quadratic auxiliary function whose minimizer is exactly the free-coordinate update in (14). The fixed hard-support zeros are not altered. Thus the V update also cannot increase J.

Alternating the two updates yields

$$J(U^{(t+1)}, V^{(t+1)}) \le J(U^{(t)}, V^{(t)}).$$

Finally, every term in (8) is non-negative, so $J \ge 0$. Therefore the non-increasing sequence of objective values converges. □

Remark 1. Theorem 2 establishes monotonic descent and convergence of the scalar objective sequence on the positive free-coordinate domain. Factor convergence and stationarity require additional assumptions. The empirical convergence study reports the finite-precision objective trajectories generated by the implemented updates.

4.6 Computational complexity

After the harmonic evidence has been computed, one PCE-NMF iteration mainly requires the products XV, $X^T U$, $V^T V$, and $U^T U$. The dominant factorization cost is therefore

$$O\big(mnc + (m+n)c^2\big)$$

per iteration. Computing the propagated-exclusion weights and their diagonal loading requires only $O(nc)$ additional operations, apart from the class-evidence propagation stage.

The harmonic propagation stage solves sparse Dirichlet systems with c right-hand sides on the anchored graph components. If $\mathcal{S}(L_{\mathcal{U}\mathcal{U}}, c)$ denotes the cost of the corresponding sparse solve, this cost is incurred once before the NMF iterations and no graph variable is updated during factorization.

Table 2: Data sets and graph neighborhood sizes. Here, p_{PCE}, p_{GOC}, and p_{SN} denote the neighborhood sizes used by PCE-NMF, GOCNMF, and SNMFWLP, respectively. GNMFLD uses $p = 5$ throughout.

Data set	Features	Samples	Classes	p_{PCE}	p_{GOC}	p_{SN}
YaleB	1024	2414	38	2	2	3
COIL20	1024	1440	20	2	3	3
Optdigits	64	5620	10	4	4	4
ORL	4096	400	40	3	3	3
PIE	1024	2856	68	3	3	3
COIL100	1024	7200	100	2	3	3

Constructing an exact cosine-neighbor graph from dense data requires $O(mn^2)$ arithmetic and $O(n^2)$ temporary storage in a direct implementation. The released code uses blocked neighbor search and stores the final graph sparsely, reducing the working memory while leaving the worst-case arithmetic order unchanged. The cost and fill-in of the component-wise Dirichlet solves depend on graph sparsity and ordering; these one-time graph operations can dominate the NMF iterations when n is large.

5 Experiments

The experiments address four questions. First, does PCE-NMF improve direct semi-supervised clustering under a paired protocol? Second, what performance is contributed by propagation beyond hard labeled support constraints? Third, how do label rate and graph neighborhood size change the result? Fourth, do the implemented updates preserve the predicted descent and support structure?

5.1 Data sets and preprocessing

Six benchmark data sets are used: YaleB [16], COIL20 [17], Optdigits [18], ORL [19], PIE [20], and COIL100 [21]. Their dimensions and graph settings are listed in Table 2. The data include face images, object images, and handwritten digits, and vary from 400 to 7200 samples and from 10 to 100 classes.

For every method, samples are represented as columns and normalized to unit Euclidean norm. No centering, dimensionality reduction, balancing, or data-set subsampling is applied. For each class k, the labeled subset contains

$$\max\{1, \lfloor 0.1 n_k \rfloor\} \tag{15}$$

randomly selected samples, where n_k is the number of samples in that class. All remaining samples are treated as unlabeled. The 10 development seeds

$$20260617, \ldots, 20260626 \tag{16}$$

are used for parameter sensitivity. The disjoint 20-seed test set

$$20260627, \ldots, 20260646 \tag{17}$$

is used for the external comparison and confirmatory ablation. Within each data-set–seed pair, all methods receive the same labeled split. Paired non-negative initial factors are also shared whenever permitted by the factorization structure.

Accuracy (ACC) and normalized mutual information (NMI) are computed only on the unlabeled index set $\mathcal{U}$. Let $\widehat{y}_i$ denote a predicted cluster index and let Π_c be the permutations of c labels. The external comparison uses

$$\mathrm{ACC} = \max_{\pi \in \Pi_c} \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \mathbf{1}\{y_i = \pi(\widehat{y}_i)\}, \tag{18}$$

$$\mathrm{NMI} = \frac{2I(Y;\widehat{Y})}{H(Y) + H(\widehat{Y})}. \tag{19}$$

The optimal permutation is an evaluation-only relabeling of the predicted partition and is computed from the unlabeled evaluation set. NMF and CNMF use 50-start K-means. Each start runs at most 100 Lloyd iterations; an empty cluster is reseeded with the sample having the largest current assignment distance, and the solution with the smallest within-cluster sum of squares is retained. GNMFLD, EDNMF, LCNMF, GOCNMF, HLPNMF, and PCE-NMF use their native row-wise assignments. SNMFWLP follows its source protocol by applying 50-start K-means to the learned low-dimensional representation. PCE-NMF predicts labels from the scale-invariant score

$$s_{ik} = \|u_k\|_2^2 v_{ik}^2, \qquad \widehat{y}_i = \arg\max_k s_{ik}. \tag{20}$$

The same permutation-aligned ACC definition is used for every method in the external table. NMI is permutation invariant. The mechanism-specific ablation and sensitivity analyses additionally report the native, unpermuted PCE-NMF decision in Eq. (9).

5.2 Compared methods and parameter protocol

The external comparison includes NMF, CNMF, GNMFLD, EDNMF, LCNMF, GOCNMF, SNMFWLP, HLPNMF, and PCE-NMF. Standard NMF provides an unsupervised reference. CNMF uses the hard constraint $V = AZ$. GNMFLD combines labeled-coordinate fitting with graph regularization [22]. The EDNMF baseline uses its element-ratio variant, ERDNMF. LCNMF suppresses the wrong-class coordinates of labeled samples. GOCNMF fixes labeled representations to one-hot vectors and uses graph smoothness. SNMFWLP combines an orthogonal basis constraint with weighted label propagation. HLPNMF constructs a hypergraph transition matrix and regularizes the representation using high-order label propagation [13].

NMF, CNMF, GNMFLD, EDNMF, and LCNMF are run for 200 multiplicative-update iterations. GOCNMF, SNMFWLP, and PCE-NMF are run for 50 iterations, and HLPNMF is run for 30 iterations as specified by its source implementation. PCE-NMF uses $\lambda = 100$ on all data sets. The value of λ and the neighborhood sizes for YaleB, COIL20, and COIL100 were selected using the 10 development seeds and fixed before evaluating the 20 test seeds. The Optdigits, ORL, and PIE neighborhood sizes were fixed by the initial locked protocol before test evaluation.

GNMFLD uses $p = 5$ throughout. PCE-NMF, GOCNMF, and SNMFWLP use the neighborhood sizes listed in Table 2; the same settings are used throughout the main comparison, ablation, and convergence experiments. Shared values are used when the verified implementations admit the same graph. EDNMF, LCNMF, GOCNMF, SNMFWLP, and HLPNMF use preserved author-code snapshots with interface adaptations for arbitrary labeled indices and paired initialization; NMF, CNMF, GNMFLD, and PCE-NMF use audited local implementations. Table 3 reports the data-set-specific parameters and their provenance.

The comparison uses configurations fixed before the independent test and a common paired evaluation protocol. Parameter provenance is marked explicitly in Table 3; parameter values, initialization modes, iteration budgets, and implementation sources are recorded in the public reproducibility files. Unlabeled ground-truth labels enter only the final ACC and NMI calculations. The parameter budget follows the source implementations. Author-code values are retained where available, and Table 3 identifies the remaining locked and transferred values.

For statistical comparison, we form paired differences between PCE-NMF and each baseline for every data-set–metric combination. A two-sided exact paired Wilcoxon signed-rank test [23] is applied to the 20 paired trials. Holm correction is performed within each pre-specified baseline family of 12 tests (six data sets times two metrics). We report the magnitude, direction, and adjusted p-value of each paired difference. Holm adjustment is computed separately for each of the eight baseline families.

All computations use MATLAB R2019a on Windows with double-precision arrays on an Intel Core i5-8300H CPU (four cores, eight logical processors) and 8 GB of memory. The reproducibility bundle records the development/test seed partition, split-generation rule, method order, implementation provenance, transferred-parameter tags, raw trial records, summaries, paired tests, and diagnostics. The formal experiment contains $6 \times 20 \times 9 = 1080$ valid independent-test method–data-set–seed runs.

5.3 Overall clustering results

Table 4 reports the mean and standard deviation over the 20 independent paired trials. Among the nine tested methods, PCE-NMF has the highest mean value on both metrics for COIL20 and COIL100 and the highest ACC on ORL. On COIL20, it reaches 0.9678 ACC and 0.9684 NMI, compared with 0.9486 and 0.9488 for HLPNMF. On COIL100, the corresponding results are 0.9093 and 0.9443, exceeding HLPNMF by 0.0324 ACC and 0.0190 NMI.

Table 5 audits the direct class semantics of PCE-NMF by comparing its native prediction in Eq. (9) with the common permutation-aligned ACC used in Table 4. The two scores are identical on YaleB, COIL20, Optdigits, and PIE. Their mean difference is 1.59×10^{-3} on COIL100 and 0.0110 on ORL. Thus, the external ACC ranking is accompanied by an explicit native class-coordinate result.

PCE-NMF exceeds GNMFLD on both metrics for YaleB, COIL20, ORL, and COIL100. The paired mean ACC improvements are 0.0806, 0.0809, 0.0236, and 0.0914, respectively. On Optdigits, PCE-NMF, GOCNMF, and GNMFLD all obtain approximately 0.984 ACC and 0.959 NMI. The PCE-NMF-GNMFLD differences are -6.62×10^{-4} ACC and -1.28×10^{-3} NMI.

GOCNMF exceeds PCE-NMF on YaleB by 0.0389 ACC and 0.0320 NMI and on PIE by 0.0070 ACC and 0.0071

Table 3: Locked data-set-specific parameters and provenance. A denotes an author-code data-set value, T a value transferred from a related source data set, and L a value fixed in the initial comparison protocol without target-test-set selection.

Data set	PCE λ	GOC α	GNMFLD α	GNMFLD β	ERDNMF α	HL (p, k)	Provenance
YaleB	100	1000	1000	50	10	$(2, 3)$ A	GOC: L; GNMFLD: L; ER: T; SN p: T
COIL20	100	10	800	200	800	$(2, 3)$ A	GOC: L; GNMFLD: L; ER: A; SN p: A
Optdigits	100	10	600	100	600	$(5, 1)$ T	GOC: L; GNMFLD: L; ER: A; SN p: T
ORL	100	100	400	10	10	$(2, 3)$ T	GOC: L; GNMFLD: L; ER: T; SN p: T
PIE	100	1000	1000	50	400	$(3, 5)$ A	GOC: A; GNMFLD: L; ER: A; SN p: T
COIL100	100	10	600	50	1000	$(2, 3)$ A	GOC: L; GNMFLD: L; ER: A; SN p: A

Table 4: Clustering performance (mean $\pm$ standard deviation) over 20 paired trials. ACC is permutation-aligned according to the definition above, and both ACC and NMI are evaluated only on unlabeled samples. Bold and underlined entries denote the best and second-best mean values, respectively.

Data set	Metric	NMF	CNMF	GNMFLD	EDNMF	LCNMF	GOCNMF	SNMFWLP	HLPNMF	PCE-NMF
YaleB	ACC	0.1582 ± 0.0088	0.1641 ± 0.0111	0.6127 ± 0.0144	0.2275 ± 0.0112	0.2614 ± 0.0127	$\underline{0.7322 \pm 0.0134}$	0.5466 ± 0.0229	$\mathbf{0.7325 \pm 0.0130}$	0.6933 ± 0.0157
	NMI	0.3012 ± 0.0115	0.3019 ± 0.0148	0.6250 ± 0.0093	0.3781 ± 0.0098	0.3840 ± 0.0094	$\underline{0.7337 \pm 0.0092}$	0.6283 ± 0.0167	$\mathbf{0.7350 \pm 0.0082}$	0.7018 ± 0.0114
COIL20	ACC	0.6276 ± 0.0194	0.6372 ± 0.0262	0.8869 ± 0.0082	0.6925 ± 0.0111	0.6886 ± 0.0125	$\underline{0.9231 \pm 0.0076}$	0.6589 ± 0.0427	$\underline{0.9486 \pm 0.0128}$	$\mathbf{0.9678 \pm 0.0084}$
	NMI	0.7368 ± 0.0082	0.7443 ± 0.0139	0.9128 ± 0.0082	0.7561 ± 0.0111	0.7498 ± 0.0112	0.9374 ± 0.0069	0.7705 ± 0.0393	$\underline{0.9488 \pm 0.0119}$	$\mathbf{0.9684 \pm 0.0047}$
Optdigits	ACC	0.7208 ± 0.0486	0.7602 ± 0.0263	$\underline{0.9840 \pm 0.0015}$	0.8257 ± 0.0105	0.7828 ± 0.0526	0.9835 ± 0.0013	0.9600 ± 0.0488	$\mathbf{0.9856 \pm 0.0014}$	0.9833 ± 0.0014
	NMI	0.6599 ± 0.0228	0.6813 ± 0.0165	$\underline{0.9592 \pm 0.0032}$	0.7143 ± 0.0082	0.6907 ± 0.0327	0.9583 ± 0.0028	0.9508 ± 0.0159	$\mathbf{0.9626 \pm 0.0026}$	0.9579 ± 0.0029
ORL	ACC	0.5775 ± 0.0252	0.5799 ± 0.0368	$\underline{0.6232 \pm 0.0280}$	0.5454 ± 0.0176	0.5906 ± 0.0183	$\underline{0.6408 \pm 0.0201}$	0.5796 ± 0.0245	0.6039 ± 0.0152	$\mathbf{0.6468 \pm 0.0218}$
	NMI	0.7714 ± 0.0129	0.7745 ± 0.0180	0.7673 ± 0.0141	0.7006 ± 0.0133	0.7365 ± 0.0120	$\underline{0.7940 \pm 0.0100}$	0.7671 ± 0.0133	$\mathbf{0.7972 \pm 0.0087}$	0.7872 ± 0.0107
PIE	ACC	0.5316 ± 0.0177	0.5643 ± 0.0218	$\underline{0.9430 \pm 0.0056}$	0.7691 ± 0.0163	0.7623 ± 0.0093	$\underline{0.9377 \pm 0.0128}$	0.5567 ± 0.0297	$\mathbf{0.9577 \pm 0.0116}$	0.9307 ± 0.0130
	NMI	0.7971 ± 0.0077	0.8048 ± 0.0083	$\underline{0.9514 \pm 0.0034}$	0.8453 ± 0.0088	0.8409 ± 0.0045	$\underline{0.9552 \pm 0.0069}$	0.7574 ± 0.0246	$\mathbf{0.9710 \pm 0.0055}$	0.9482 ± 0.0066
COIL100	ACC	0.4631 ± 0.0104	0.4820 ± 0.0139	0.8179 ± 0.0075	0.5516 ± 0.0100	0.5424 ± 0.0079	$\underline{0.8520 \pm 0.0075}$	0.4071 ± 0.0142	$\underline{0.8769 \pm 0.0105}$	$\mathbf{0.9093 \pm 0.0051}$
	NMI	0.7393 ± 0.0061	0.7465 ± 0.0038	0.8947 ± 0.0034	0.7224 ± 0.0055	0.7183 ± 0.0041	0.9140 ± 0.0036	0.6231 ± 0.0150	$\underline{0.9253 \pm 0.0055}$	$\mathbf{0.9443 \pm 0.0025}$

Table 5: PCE-NMF ACC under permutation-aligned and native class-semantic evaluation. Values are mean $\pm$ standard deviation over the same 20 trials and include unlabeled samples only.

Data set	Aligned ACC	Native direct ACC
YaleB	0.6933 ± 0.0157	0.6933 ± 0.0157
COIL20	0.9678 ± 0.0084	0.9678 ± 0.0084
Optdigits	0.9833 ± 0.0014	0.9833 ± 0.0014
ORL	0.6468 ± 0.0218	0.6358 ± 0.0244
PIE	0.9307 ± 0.0130	0.9307 ± 0.0130
COIL100	0.9093 ± 0.0051	0.9077 ± 0.0051

NMI. On ORL, PCE-NMF improves ACC by 0.0060, while GOCNMF improves NMI by 0.0067.

HLPNMF attains the highest mean ACC and NMI on YaleB, Optdigits, and PIE. PCE-NMF exceeds HLPNMF by $0.0192/0.0196$ ACC/NMI on COIL20 and by $0.0324/0.0190$ on COIL100. ORL separates the two criteria: PCE-NMF improves ACC by 0.0429, whereas HLPNMF improves NMI by 0.0100. All 12 paired comparisons remain significant after Holm correction.

Against NMF, CNMF, EDNMF, and LCNMF, PCE-NMF achieves higher mean ACC and NMI on all six data sets, with all corresponding Holm-adjusted tests significant. PCE-NMF also exceeds SNMFWLP in mean performance on all six data sets after SNMFWLP is evaluated by 50-start K-means on its learned representation, as specified by its source protocol. The Holm-adjusted paired tests are significant on YaleB, COIL20, ORL, PIE, and COIL100. On Optdigits, the small mean differences are not significant after Holm correction.

5.4 Paired comparison with the strongest baselines

Table 6 compares PCE-NMF with GNMFLD, GOCNMF, and HLPNMF. Differences are PCE-NMF minus the stated baseline.

Figure 2 visualizes the paired comparisons with GNMFLD and GOCNMF using two-sided 95% Student confidence intervals for the mean difference across the 20 matched seeds. The zero line separates improvements from degradations. Gains over GNMFLD are largest on YaleB and COIL100. Differences among PCE-NMF, GNMFLD, and GOCNMF are close to zero on Optdigits, while ORL exhibits opposite ACC and NMI directions relative to GOCNMF.

On Optdigits, the differences between PCE-NMF and GOCNMF are below 5×10^{-4}, while HLPNMF has the highest mean ACC and NMI. On ORL, PCE-NMF attains the highest ACC, whereas HLPNMF attains the highest NMI.

5.5 Parameter sensitivity

We perform a sensitivity analysis on YaleB, COIL20, and COIL100 using the 10 development seeds. We vary $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$ at the locked neighborhood size, and vary $p \in \{2, 3, 4, 5, 6\}$ at $\lambda = 100$. All other splits, initializations, updates, and unlabeled-only evaluations are unchanged. Figure 3 reports native PCE-NMF ACC and NMI; open circles mark the settings subsequently fixed for the 20-seed test comparison.

Performance increases sharply between $\lambda = 0.01$ and

Table 6: Paired comparisons with GNMFLD, GOCNMF, and HLPNMF. Values are mean paired differences; parentheses contain Holm-adjusted p-values.

	PCE−GNMFLD		PCE−GOCNMF		PCE−HLPNMF	
Data set	ACC	NMI	ACC	NMI	ACC	NMI
YaleB	$+0.08059\ (2.29\times10^{-5})$	$+0.07678\ (2.29\times10^{-5})$	$-0.03891\ (2.29\times10^{-5})$	$-0.03198\ (2.29\times10^{-5})$	$-0.03921\ (2.29\times10^{-5})$	$-0.03322\ (2.29\times10^{-5})$
COIL20	$+0.08092\ (2.29\times10^{-5})$	$+0.05557\ (2.29\times10^{-5})$	$+0.04473\ (2.29\times10^{-5})$	$+0.03105\ (2.29\times10^{-5})$	$+0.01923\ (2.29\times10^{-5})$	$+0.01963\ (2.29\times10^{-5})$
Optdigits	$-0.00066\ (0.00245)$	$-0.00128\ (0.00245)$	$-0.00017\ (0.265)$	$-0.00037\ (0.265)$	$-0.00227\ (2.29\times10^{-5})$	$-0.00472\ (2.29\times10^{-5})$
ORL	$+0.02361\ (3.43\times10^{-4})$	$+0.01999\ (1.60\times10^{-4})$	$+0.00597\ (0.0573)$	$-0.00672\ (6.71\times10^{-4})$	$+0.04292\ (2.29\times10^{-5})$	$-0.00997\ (0.00831)$
PIE	$-0.01231\ (1.60\times10^{-4})$	$-0.00321\ (0.0441)$	$-0.00700\ (2.29\times10^{-5})$	$-0.00705\ (2.29\times10^{-5})$	$-0.02703\ (2.29\times10^{-5})$	$-0.02287\ (2.29\times10^{-5})$
COIL100	$+0.09136\ (2.29\times10^{-5})$	$+0.04963\ (2.29\times10^{-5})$	$+0.05735\ (2.29\times10^{-5})$	$+0.03033\ (2.29\times10^{-5})$	$+0.03240\ (2.29\times10^{-5})$	$+0.01904\ (2.29\times10^{-5})$

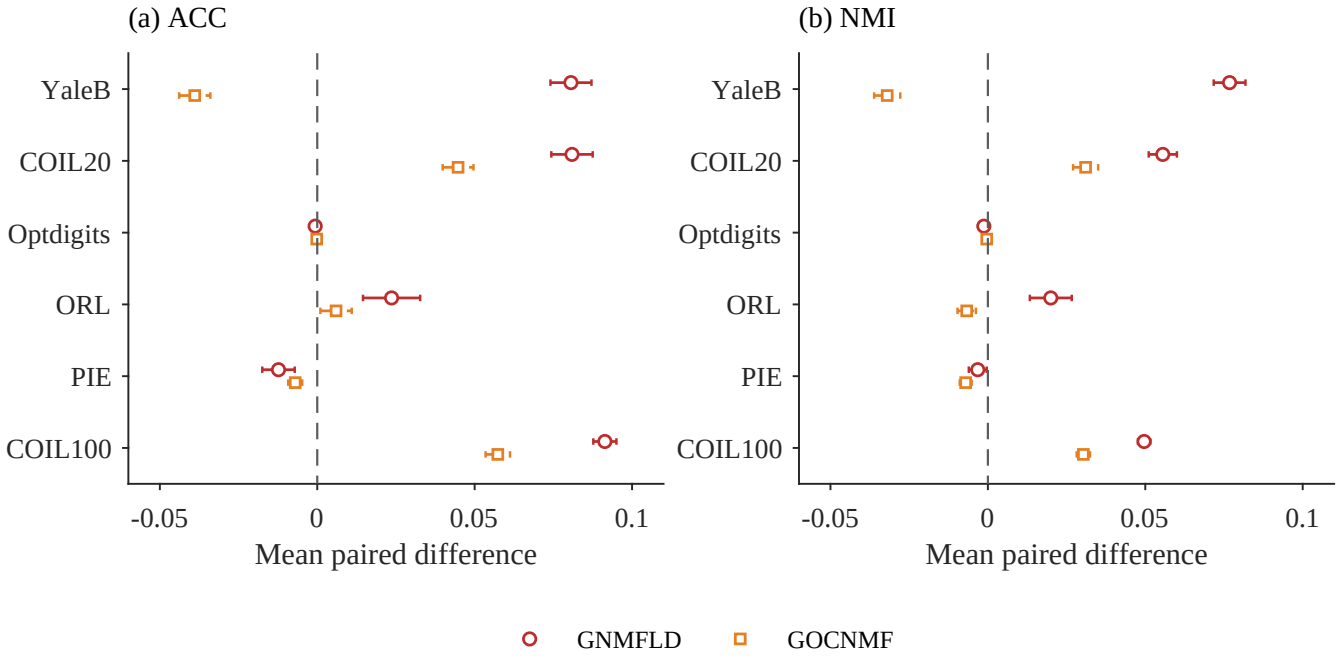


Figure 2: Paired performance differences of PCE-NMF relative to GNMFLD and GOCNMF. For baseline B and metric M, let $\Delta_B = M(\text{PCE-NMF}) - M(B)$, where $B \in \{\text{GNMFLD}, \text{GOCNMF}\}$ and $M \in \{\text{ACC}, \text{NMI}\}$. Panels (a) and (b) report ACC and NMI, respectively. Markers and error bars show the mean and the interval $\overline{\Delta}_B \pm t_{0.975,19} s_\Delta / \sqrt{20}$ over the matched seeds. The dashed line marks $\Delta_B = 0$, and positive values favor PCE-NMF.

10, and changes only slightly from $\lambda = 100$ to 1000. At $\lambda = 100$, the mean direct ACC values are 0.7000, 0.9710, and 0.9104 on YaleB, COIL20, and COIL100; at $\lambda = 1000$ they are 0.7017, 0.9709, and 0.9114. The setting $\lambda = 100$ therefore lies on a broad performance plateau.

The neighborhood size is more influential. All three data sets attain their highest tested mean ACC at $p = 2$. Relative to $p = 3$, direct ACC increases from 0.9235 to 0.9712 on COIL20 and from 0.8665 to 0.9103 on COIL100. The 20-seed comparison uses $p = 2$ for YaleB, COIL20, and COIL100.

5.6 Influence of supervisory information

Because PCE-NMF is designed to propagate limited class evidence, we perform a label-rate robustness analysis under different amounts of supervision. YaleB, COIL20, and COIL100 provide one face data set and two object data sets with 20 and 100 classes, respectively, and therefore expose both data-type and class-count changes within a tractable repeated experiment. The labeled proportion per class is varied over $\{5\%, 10\%, 15\%, 20\%\}$ using nested labeled sets, while a common evaluation pool is retained for all proportions within each paired trial. Figure 4 reports the mean ACC and NMI over 20 paired seeds. SNMFWLP is evaluated by applying 50-start K-means to its learned representation in every trial, consistent with its native prediction protocol.

The benefit of propagated class exclusion is most pronounced in the low-supervision regime. On COIL20 with only 5% labeled samples, PCE-NMF achieves an ACC of 0.9446, compared with 0.8274 for GOCNMF, while the corresponding NMI values are 0.9555 and 0.8620. The advantage is even more evident on COIL100, where PCE-NMF improves ACC from 0.7063 to 0.8602 and NMI from 0.8312 to 0.9210.

PCE-NMF remains ahead of GOCNMF across the four tested supervision levels on COIL20 and COIL100. Its performance increases with the labeled proportion, while the largest absolute gains over GOCNMF occur in the low-supervision regime.

On YaleB, GOCNMF remains stronger over all four supervision levels, and PCE-NMF consistently outperforms GNMFLD and SNMFWLP. The strongest PCE-NMF gains in this analysis occur in low-supervision object-clustering problems with many classes.

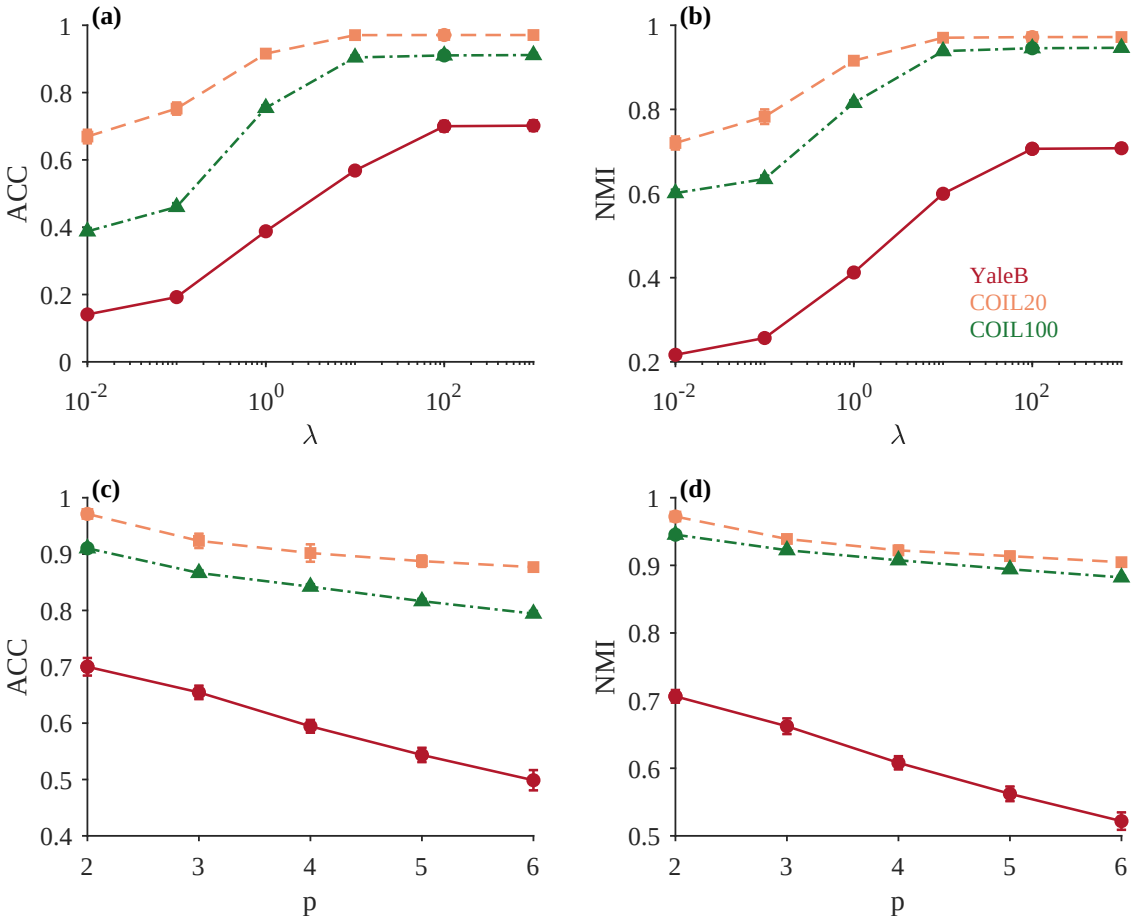


Figure 3: Parameter sensitivity of PCE-NMF over 10 paired seeds. (a,b) ACC and NMI versus λ at the locked p; (c,d) ACC and NMI versus p at $\lambda = 100$. Error bars show one standard deviation and open circles mark the settings fixed before the 20-seed test comparison.

5.7 Ablation of propagated class exclusion

To isolate the contribution of propagated class exclusion, we compare PCE-NMF with a hard-class-exclusion variant, denoted HCE-NMF. HCE-NMF preserves the same factorization structure and the labeled wrong-class zero constraints but sets the propagated-exclusion regularization weight to zero. This comparison measures the incremental effect of the complete propagated-exclusion term over the hard-support model.

Both variants have class-semantic latent coordinates and are evaluated using the native class-energy decision

$$\widehat{y}_i = \arg\max_k \|u_k\|_2^2 v_{ik}^2, \tag{21}$$

without permutation alignment. Both variants use the same seeds, labeled splits, initial factors, cosine graphs, updates, and NMI definition. Table 7 reports the ablation results over the 20 paired trials.

PCE-NMF improves both metrics on every data set. The mean ACC gains over HCE-NMF range from 0.3097 on COIL20 to 0.5925 on PIE, while the NMI gains range from 0.2552 to 0.4893. PCE-NMF wins all 120 seed-level comparisons for each metric. All 12 data-set–metric tests remain significant after Holm correction (adjusted $p = 2.29\times10^{-5}$). These results show that hard compatibility constraints alone are insufficient in this model: propagation supplies useful class-incompatibility information for unlabeled samples.

To separate this gain from the predictive strength of harmonic propagation itself, we also evaluate a propagation-only control. It uses the same graph, labeled split, and component-wise harmonic field as PCE-NMF and predicts

$$\widehat{y}_i^{\mathrm{LP}} = \arg\max_k P_{ik}. \tag{22}$$

For a connected component without a labeled anchor, all entries of $P_{i:}$ are zero; such samples are assigned to the most frequent class in the labeled set, with ties resolved by the smallest class index. The evaluation set is therefore identical for LP-only and PCE-NMF.

PCE-NMF improves both metrics over LP-only on COIL20 and COIL100, with mean gains of $0.0133/0.0118$ and $0.0228/0.0117$ in ACC/NMI, respectively. It also improves ACC on ORL by 0.0122, while LP-only has 0.0040 higher NMI. LP-only is stronger on YaleB and PIE. On Optdigits, the absolute differences are 4.0×10^{-4} ACC and 9.2×10^{-4} NMI. All 12 paired tests remain significant after Holm correction; the adjusted values are 7.84×10^{-4} for both

Table 7: Ablation of propagated class exclusion under the native direct class-energy decision. Values are mean ± standard deviation over 20 paired trials.

	HCE-NMF		PCE-NMF	
Data set	ACC	NMI	ACC	NMI
YaleB	0.1331 ± 0.0119	0.2125 ± 0.0129	0.6933 ± 0.0157	0.7018 ± 0.0114
COIL20	0.6582 ± 0.0144	0.7132 ± 0.0132	0.9678 ± 0.0084	0.9684 ± 0.0047
Optdigits	0.6538 ± 0.0930	0.6090 ± 0.0490	0.9833 ± 0.0014	0.9579 ± 0.0029
ORL	0.2769 ± 0.0275	0.5129 ± 0.0195	0.6358 ± 0.0244	0.7872 ± 0.0107
PIE	0.3382 ± 0.0191	0.5081 ± 0.0170	0.9307 ± 0.0130	0.9482 ± 0.0066
COIL100	0.3735 ± 0.0094	0.5978 ± 0.0066	0.9077 ± 0.0051	0.9443 ± 0.0025

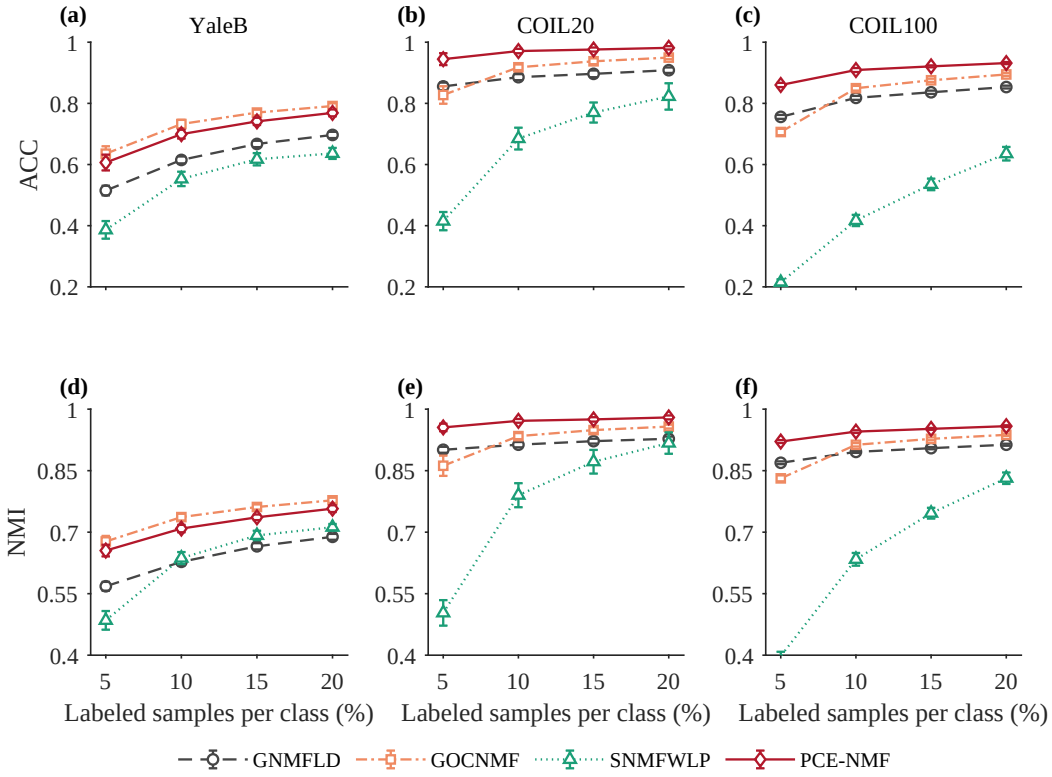


Figure 4: Influence of the proportion of labeled samples on clustering performance. The upper row reports ACC and the lower row reports NMI on YaleB, COIL20, and COIL100. For each paired seed, the labeled sets are nested across 5%, 10%, 15%, and 20% supervision, while all settings share a common evaluation pool. Markers denote the mean results over the separate 20-seed robustness analysis (20260617–20260636), and error bars indicate one standard deviation. SNMFWLP uses 50-start K-means on its learned representation.

Table 8: Harmonic propagation alone and PCE-NMF under native class-semantic decisions. Values are mean ± standard deviation over the 20 independent test seeds.

	Harmonic LP-only		PCE-NMF	
Data set	ACC	NMI	ACC	NMI
YaleB	0.7365 ± 0.0125	0.7384 ± 0.0086	0.6933 ± 0.0157	0.7018 ± 0.0114
COIL20	0.9546 ± 0.0099	0.9566 ± 0.0068	0.9678 ± 0.0084	0.9684 ± 0.0047
Optdigits	0.9837 ± 0.0013	0.9588 ± 0.0028	0.9833 ± 0.0014	0.9579 ± 0.0029
ORL	0.6236 ± 0.0245	0.7913 ± 0.0091	0.6358 ± 0.0244	0.7872 ± 0.0107
PIE	0.9488 ± 0.0124	0.9628 ± 0.0060	0.9307 ± 0.0130	0.9482 ± 0.0066
COIL100	0.8849 ± 0.0079	0.9326 ± 0.0039	0.9077 ± 0.0051	0.9443 ± 0.0025

Optdigits metrics and at most this value for the other comparisons. Harmonic evidence is already a strong classifier, while its coupling to contribution-energy shrinkage supplies the additional gains on COIL20 and COIL100.

Figure 5 traces this mechanism in a formal COIL20 trial. The harmonic evidence concentrates near the correct class coordinate, its complement assigns small exclusion weight to that coordinate and large weights to incompatible coordinates, and the learned class-energy share becomes strongly diagonal. Thus the observed gain is accompanied by the matrix pattern predicted by the PCE penalty.

5.8 Sample-level visualization of propagated exclusion

Figure 6 presents three sample-level mechanism cases from the propagation-to-decision path on ORL. The samples are selected deterministically by scanning the true classes and sample indices and retaining the first sample from each of the first three classes for which HCE-NMF is incorrect and PCE-NMF is correct. All quantities come from the same labeled split, initialization, graph, and seed used by the paired ablation. The harmonic evidence $P_{i:}$ identifies compatible class coordinates, and $Q_{i:} = \mathbf{1} - P_{i:}$ assigns larger exclusion weights to incompatible coordinates. The final two columns show the normalized class-contribution energies

$$\bar{s}_{ik} = \frac{\|u_k\|_2^2 v_{ik}^2}{\sum_j \|u_j\|_2^2 v_{ij}^2}. \tag{23}$$

For these cases, HCE-NMF distributes substantial energy over several incompatible coordinates and assigns the largest energy to an incorrect class. Propagated exclusion suppresses these competing coordinates and moves the largest normalized contribution energy to the true class. This sample-level transition illustrates the mechanism, while the paired ablation in Table 7 provides the corresponding performance evidence over all seeds.

5.9 Visualization of learned representations

We visualize the scale-invariant representations learned on COIL100, where PCE-NMF ranks first in both ACC and NMI in the formal comparison. For sample i, the representation is

$$z_{ik} = \|u_k\|_2 v_{ik}, \qquad k = 1, \ldots, c, \tag{24}$$

so that z_{ik}^2 equals the class-wise contribution energy used by the PCE-NMF decision rule. This representation is unchanged by the standard NMF diagonal scaling transformation.

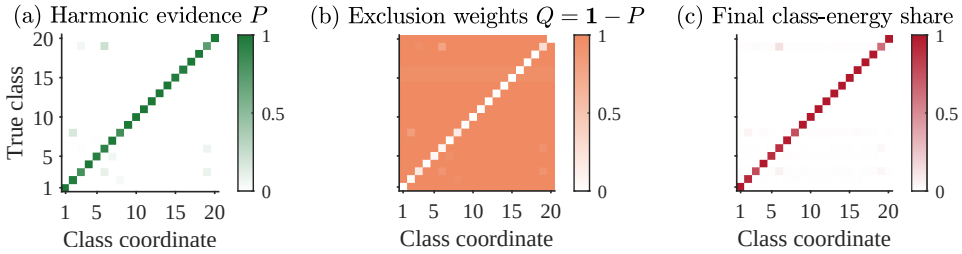
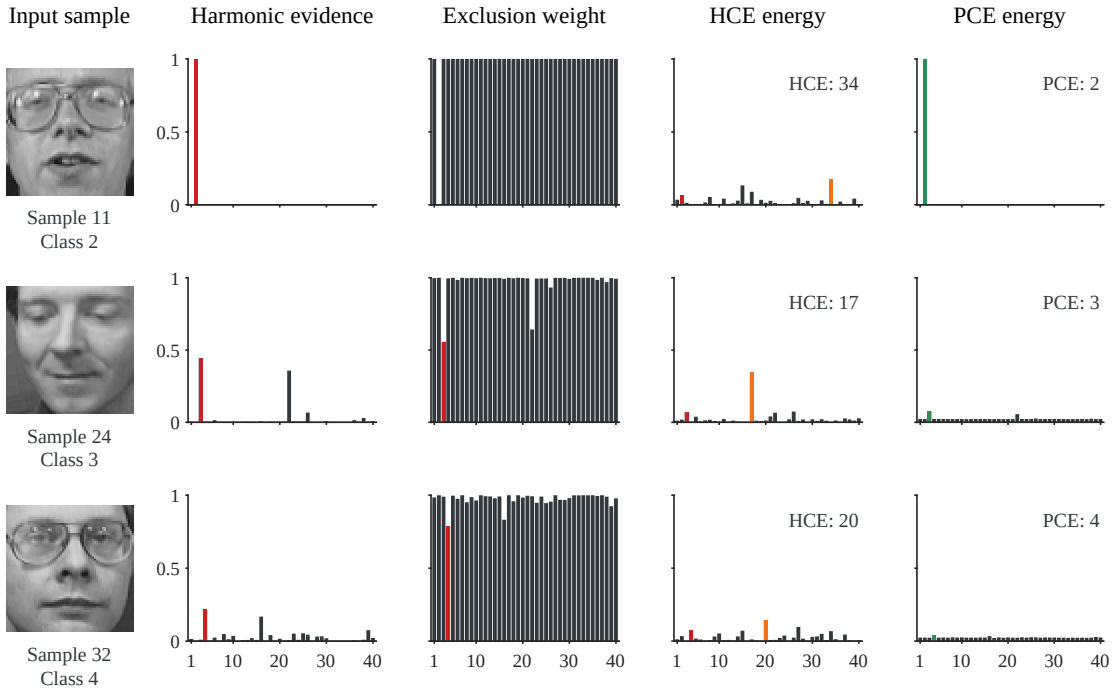


Figure 5: Propagation-to-factorization mechanism on COIL20 for development seed 20260617, $p = 2$, $\lambda = 100$, and 50 iterations. Each row averages all unlabeled samples of one true class. (a) Harmonic class evidence P; (b) propagated exclusion weights $Q = 1 - P$; and (c) the final normalized class-energy share $s_{ik} / \sum_j s_{ij}$.



ORL, seed 20260617. Red: true class; orange: incorrect HCE prediction; green: correct PCE prediction.

Figure 6: Three sample-level cases of the propagated-exclusion mechanism on ORL for development seed 20260617, $p = 3$, $\lambda = 100$, and 50 iterations. The rows show samples 11, 24, and 32, whose true classes are 2, 3, and 4. From left to right: input sample, harmonic class evidence, propagated exclusion weight, normalized HCE-NMF class energy, and normalized PCE-NMF class energy. Red marks the true class coordinate, orange marks the incorrect HCE-NMF decision, and green marks the correct PCE-NMF decision. HCE-NMF predicts classes 34, 17, and 20, whereas PCE-NMF predicts classes 2, 3, and 4.

Figure 7 shows the local structure within each learned representation. Each t-SNE panel has an independent orientation, scale, and coordinate system. Table 4 reports ACC and NMI computed from the original representations.

5.10 Empirical convergence behavior

We record the PCE-NMF objective for 50 iterations on YaleB, COIL20, and COIL100 using the 10 development seeds and the formal parameters. Figure 8 shows the mean normalized objective $F^{(t)}/F^{(1)}$.

All 30 recorded trajectories are finite and nonincreasing under the numerical tolerance used by the implementation. At iteration 50, the mean ratios $F^{(50)}/F^{(1)}$ are 0.9276, 0.3051, and 0.3160 on YaleB, COIL20, and COIL100, respectively. The corresponding mean relative changes in the final step are 1.60×10^{-7}, 2.61×10^{-5}, and 1.74×10^{-4}. These measurements establish objective-value descent for the recorded runs.

5.11 Propagation coverage and constraint diagnostics

The harmonic field is defined component-wise, and propagated class-exclusion weights are introduced only on connected components that contain at least one labeled anchor. An unanchored component contains no label information for class-evidence inference, so PCE-NMF assigns zero propagated-exclusion weight to that component.

Across the 20 independent test seeds, the mean anchor coverage is 1.000 on YaleB, 0.985 on COIL20, 1.000 on Optdigits, 0.940 on ORL, 0.994 on PIE, and 0.962 on COIL100. The corresponding minimum coverage values are 1.000, 0.975, 1.000, 0.919, 0.979, and 0.944, respectively. Thus, all unlabeled samples are anchored on YaleB and Optdigits, while only a small fraction of samples belongs to u-

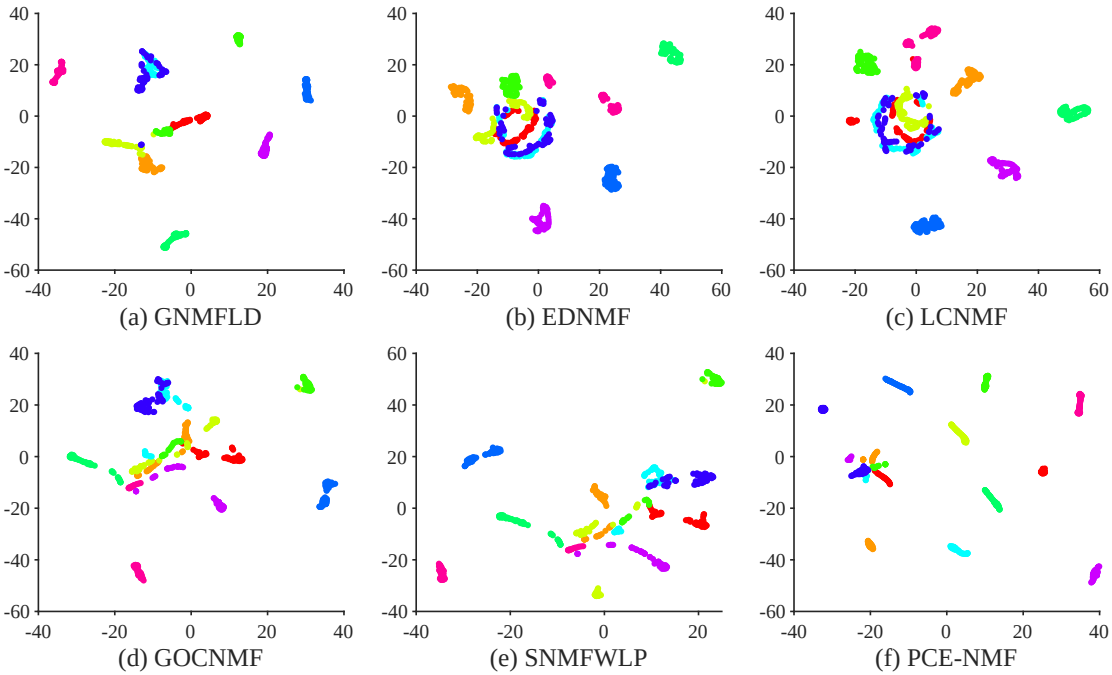


Figure 7: Two-dimensional visualization of scale-invariant representations for ten prespecified COIL100 classes: (a) GNMFLD, (b) EDNMF, (c) LCNMF, (d) GOCNMF, (e) SNMFWLP, and (f) PCE-NMF. The plots use the same unlabeled samples, class colors, t-SNE configuration, and development seed 20260617. Each representation is mapped independently using Euclidean distance, perplexity 30, and no input standardization. Each independently fitted panel has its own coordinate system and provides a qualitative within-panel view.

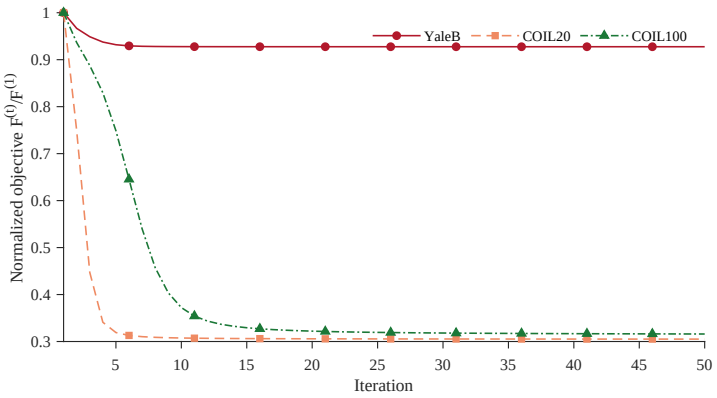


Figure 8: Mean objective trajectories of PCE-NMF over the 10 development seeds at the formal settings. Curves show $F^{(t)}/F^{(1)}$; sparse markers distinguish the three data sets.

nanchored components on the other four data sets. The minimum observed coverage remains above 0.91 on every data set. These results show that the harmonic evidence reaches the great majority of unlabeled samples under the graph settings used in the formal experiments, while the componentwise construction prevents unsupported class evidence from being assigned to disconnected regions.

We further examine numerical feasibility of the constrained factorization. All 1080 method runs return finite factors and finite evaluation metrics. For PCE-NMF, the labeled wrong-class entries are reset to zero after every multiplicative update, and the recorded support-violation field is zero in all runs. The same holds for LCNMF, which also imposes hard wrong-class support constraints. Hence, the implemented updates preserve the prescribed support structure. Together, the coverage and feasibility diagnostics confirm that the propagated evidence is applied only where labeled information is available and that the hard exclusion constraints are maintained consistently during optimization.

5.12 Discussion

The experiments separate two sources of supervision. Hard support constraints assign class semantics to labeled coordinates, while propagated exclusion extends class-incompatibility information to unlabeled samples. Removing the second source produces large losses on every data set and every paired seed. The diagonal patterns in Fig. 5 connect this performance change to the intended sequence from harmonic evidence to exclusion weights and class-energy

concentration.

The external comparison shows complementary strengths among PCE-NMF, GOCNMF, and HLPNMF. HLPNMF leads on YaleB, Optdigits, and PIE; PCE-NMF leads on both metrics for COIL20 and COIL100 and on ACC for ORL; HLPNMF leads on ORL NMI. The comparison therefore separates regimes favored by harmonic exclusion, one-hot graph smoothing, and high-order hypergraph propagation. The label-rate analysis shows the largest PCE-NMF gains on the multi-class object data at the lowest examined label rate, with smaller gaps as direct supervision increases.

The external ranking uses the same permutation-aligned ACC definition for all methods. The native PCE-NMF decision in Eq. (9) gives the same mean ACC on YaleB, COIL20, Optdigits, and PIE; the mean absolute gap is 1.59×10^{-3} on COIL100 and 0.0110 on ORL. Hence the class-semantic coordinates support direct prediction while the aligned score preserves a common clustering comparison.

The LP-only control clarifies what is supplied by the harmonic field and what is supplied by its coupling with NMF. Harmonic propagation alone is strongest on YaleB and PIE and is effectively tied with PCE-NMF on Optdigits. PCE-NMF adds clear gains on COIL20 and COIL100 and improves ORL ACC. The factorization therefore contributes most on the two multi-class object data sets, whereas the propagated evidence itself accounts for much of the performance on the face and digit data. The proposed penalty therefore contributes a distinct factorization gain on the two multi-class object data sets.

Graph construction controls how far class evidence travels. The λ curves place $\lambda = 100$ on a broad plateau, whereas the neighborhood curves show large changes from $p = 3$ to $p = 2$ on COIL20 and COIL100. Thus, p is the principal graph parameter. The neighborhood values in Table 2 are used consistently in the main comparison, ablation, and label-rate robustness analysis.

The objective trajectories, anchor-coverage measurements, and zero-support diagnostics connect the mathematical model to the executed algorithm. They show monotonic finite-precision descent in all recorded trajectories, broad reach of the harmonic evidence, and exact preservation of labeled wrong-class zeros.

The data-set pattern distinguishes the regimes favored by the strongest mechanisms. HLPNMF leads on the YaleB and PIE face data and on Optdigits, whereas PCE-NMF shows its largest gains on COIL20 and COIL100 under low supervision. Across the three label-rate data sets, this advantage decreases as direct supervision increases. Neighborhood size has a larger effect than λ on COIL20 and COIL100, establishing graph calibration as the principal model-selection step. The blocked neighbor search and sparse harmonic solve provide the computational route for extending PCE-NMF beyond the largest collection tested here ($n = 7200$).

6 Conclusion

PCE-NMF converts harmonic class evidence into a class-incompatibility field and applies it to diagonal-scale-invariant rank-one contribution energies. The same energies define a direct class decision, linking supervision, factorization, and assignment through one mathematical quantity. The canonical representation expresses this mechanism as sample- and class-specific quadratic shrinkage, and the ideal alternating updates generate a non-increasing objective sequence on the stated positive domain. Across six data sets and nine tested methods, PCE-NMF achieves the highest mean ACC and NMI on COIL20 and COIL100 and the highest ACC on ORL. HLPNMF leads on YaleB, Optdigits, and PIE, and on ORL NMI. The HCE-NMF ablation establishes the value of propagated exclusion beyond hard class support. The LP-only comparison further shows where factorization adds value: PCE-NMF improves both metrics on COIL20 and COIL100, while harmonic propagation alone is stronger on YaleB and PIE and is nearly equal on Optdigits. The main empirical benefit is therefore concentrated in low-supervision multi-class object clustering. Propagated class exclusion provides a direct route from partial labels to class-semantic NMF clustering and a scale-invariant mechanism for coupling propagation with non-negative factorization.

CRediT authorship contribution statement

Deshu Sun: Writing–original draft, Methodology, Software, Conceptualization, Formal analysis.
Feng Wang: Writing–review and editing, Funding acquisition, Visualization, Conceptualization. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PCE-NMF solver, formal protocol, independent-test trial records, analysis scripts, and figure source data are available at `https://github.com/matrix-research-lab-Wf/PCE-NMF`. The six benchmark data sets are available from the original providers cited in Section 5.1; their redistribution terms are retained. The repository distributes the PCE-NMF implementation. Third-party baseline implementations are obtained from their respective authors or official repositories.

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